

Response to the reviewers

Manuscript: *Minimum Weighted Hazard-Exposure Dispatch: A Scheduling View with Verified Algorithms and a Wildfire Case Study* (formerly *Minimum Weighted Hazard-Exposure Dispatch: Complexity, an Exact Algorithm, and an FPTAS*)
Journal of Combinatorial Optimization — major revision

Dear Editor and Reviewers,

Thank you for the careful reading and the constructive reports. We revised the manuscript substantially. Reviewer 1’s central objection, that the theoretical results follow from the classical theory of $1\|\sum w_j U_j$, is correct, and we have rewritten the title, abstract, introduction, related work and summary table to say so plainly, to cite the classical source for each result, and to separate what is classical from the few things we believe are new. Reviewer 1’s question about the FPTAS exposed a real weakness in our experiments (the FPTAS’s rounding was inactive in most of them), which we explain and fix. Reviewer 2’s requests for a clearer hardness proof, a self-contained proof of the equal-cost theorem, a shorter positioning section and a discussion of parameter uncertainty are all addressed.

Before resubmitting we re-checked every numerical result, every cited source, and the proofs (the statements and proofs are now also machine-checked in Lean 4), and we had the revision critically reviewed internally. This found several errors of our own, some in the first version and some introduced while revising; we list all of them in Section 3. The most consequential are: the case study had used extrapolated hazard times that the cited NIST report in fact documents (and that were far off), and had a modelling premise that its own geography contradicts; an extension we had presented as new (the equal-cost theorem for several vehicles) is classical; and our “tight” FPTAS guarantee was not the best provable one. All tables and figures were regenerated.

Numbering. Items are numbered in order of appearance, so the numbers of the submitted version have shifted. Submitted $\rightarrow$ now: Lemma 1 $\rightarrow$ Lemma 2; Theorem 2 (hardness) $\rightarrow$ Theorem 3; Theorem 3 (dynamic program) $\rightarrow$ Theorem 5; Theorem 4 (FPTAS) $\rightarrow$ Theorem 7; Theorem 5 (equal cost) $\rightarrow$ Theorem 8; Propositions 6, 7, 8 $\rightarrow$ Propositions 9, 10, 11; Theorem 9 (classification) $\rightarrow$ Theorem 12. New: Proposition 1 (the round-trip model is conservative), Proposition 4 (constant hazard-arrival time), Lemma 6 (value-indexed table). Fact 1 keeps its number. Below, “Theorem a (b)” means Theorem a now and Theorem b in the submitted version. Section numbers: Section 5 now has a new Subsection 5.5 (general-purpose solver), so robustness is 5.6 and the Camp Fire study is 5.7.

Summary of the main changes

1. **Positioning and novelty** (title, abstract, Sections 1, 2, Table 2, conclusion): every result that is classical is labelled as such with its source (Karp 1972; Lawler and Moore 1969; Sahni 1976; Gens and Levner 1981; Lawler 1976; Dessouky et al. 1990; Moore 1968), and the claims to novelty are cut down to what we can support.
2. **What we now claim** (Section 4 and Table 2): a refined, exactly tight analysis of the FPTAS (Theorem 7, Proposition 11); hardness under the arrival reading even when the hazard-arrival time is constant (Proposition 4); the observation that the depot round-trip model is conservative

with respect to a chained route (Proposition 1); and elementary worst-case examples for natural heuristics (Propositions 9 and 10). The equal-cost theorem (Theorem 8) now covers vehicles with different speeds and is labelled classical; the classification (Theorem 12) is labelled an elementary corollary.

3. **Hardness proof** (Theorem 3 (2)): a direct reduction from PARTITION with an explicit lemma (Fact 1) on feasibility under a common deadline.
4. **Experiments** (Section 5): redesigned, with exhaustive-search verification, instance sizes to $n = 400$, a stronger baseline, FPTAS experiments in which rounding is active, an adversarial family, and a comparison with a general-purpose MIP solver.
5. **Case study** (Section 5.7): rebuilt on hazard-arrival times documented by NIST for every community (nothing is extrapolated), with a comparison against a chained-route model, a sensitivity analysis with a stated protocol, and a sample-average plan.
6. **Discussion** (Section 6): corrected multi-vehicle subsection; parameter uncertainty in a new subsection with the two suggested references.
7. **Reproducibility**: all code and a Lean 4 formalisation of the theory are public (Data Availability).

1 Reviewer 1

While the manuscript is technically correct and generally well presented, the level of novelty of the theoretical contributions is not sufficiently convincing. [...] most of the main theoretical results appear to follow from well-known results in the scheduling literature. [...] Consequently, the manuscript does not clearly establish new complexity results, new algorithmic techniques, or substantially new structural properties beyond those already known for the equivalent scheduling formulation.

Response. We agree. The manuscript proves that MWHED is $1 \parallel \sum w_j U_j$ and the general results follow from it. We have not tried to argue otherwise. Concretely:

- The title, abstract and Section 1 now say that the basic complexity picture and the algorithms are inherited from scheduling theory and that we claim no new technique for the general problem. The contributions are split into a list of classical results restated with complete proofs (hardness, the dynamic program, the FPTAS, the equal-size greedy, the classification) and a list of modest additions.
- Section 2 states, for each result, which classical result it is: Theorem 5 (3) is the Lawler–Moore dynamic program; Theorem 7 (4) is the classical value-scaling scheme for job sequencing with deadlines (Sahni 1976, Gens and Levner 1981); the hardness is Karp’s (Karp, 1972) and the knapsack equivalence; and Theorem 8 (5) is the scheduling of unit-time jobs on identical or uniform machines, for which $O(n \log n)$ algorithms are known (Lawler, 1976; Dessouky et al., 1990). Our submitted version presented the extension of Theorem 8 to several vehicles as new; it is not, and we now say so (see Section 3).
- The summary table (Table 2) has a column giving the origin of each result (classical, Lawler–Moore, Moore, elementary corollary, new).

- We removed the claim that the dynamic program and the FPTAS are “necessary” for any guarantee: Propositions 9 and 10 show only that deadline-order dispatch and one greedy repair have no guarantee.

1. *Could the authors identify a new structural property?*

Response. Yes, but they are modest, and we present them as such.

- **Proposition 4 (constant hazard-arrival time).** Under the arrival reading of Remark 1 the natural homogeneous case is a hazard that reaches all sites at the same time, which is *not* a constant deadline d_i (it is $d_i = H + p_i/2$). We prove that MWHED stays weakly NP-hard there, even with all p_i even and $w_i = p_i/2$; constant p or constant w remain polynomial, so the restriction cannot be sharpened.
- **Proposition 1 (the round-trip model is conservative).** If the sites lie in a metric space, a crew that chains the sites reaches each site no later than in the depot round-trip model, so the round-trip optimum is a lower bound on what a crew can protect. This makes explicit when the premise of the model is costly (clustered sites) and is quantified on our case study.
- **A sharper, exactly tight FPTAS analysis** (Theorem 7, Proposition 11): the scheme of Theorem 7 guarantees a constant $\rho_{n,\epsilon}$ that exceeds both $1 - \epsilon$ and $1/(1 + \epsilon)$, and the constant is attained (two extremal families), so the classical $1 - \epsilon$ bound is not attained.
- **The classification** (Theorem 12) remains, but we now call it an elementary corollary, and we note that the natural tractability boundary in the scheduling literature is Lawler’s agreeable-weights case (Lawler, 1976), which contains both constant p and constant w .

2. *The manuscript repeatedly states the dynamic program and FPTAS as major contributions. However, the relationship between these algorithms and classical Lawler-Moore-type methods is not discussed in sufficient detail. Are the algorithmic contributions truly different from known scheduling algorithms?*

Response. No. Theorem 5’s dynamic program is the Lawler–Moore dynamic program (jobs in due-date order, state equal to the accepted processing time, one test per accepted job); Remark 2 says so and describes exactly what our proof adds (a derivation from the earliest-deadline-first lemma in the notation of the dispatch problem). Theorem 7 is the classical value-scaling scheme of Sahni (1976) and Gens and Levner (1981); the manuscript states this, notes that faster schemes are known, and keeps the classical construction because its analysis is the cleanest and, with Theorem 7 and Proposition 11, exactly tight. Neither algorithm is presented as a contribution; they are a self-contained treatment for a transportation audience.

3. *Why does $FPTAS = 1000$ always appears in every table? Could the authors strengthen the empirical evaluation?*

Response. This was a genuine weakness. With weights at most 100 the scaling factor $K = \max(1, \epsilon w_{\max}/n)$ equals 1 in almost all of our runs, so no rounding took place and the “FPTAS”

coincided with the exact value-indexed dynamic program. A ratio of 1.000 was therefore guaranteed and said nothing about the scheme. We state this explicitly (Section 5.2, last row of Table 4: the share of instances with $K = 1$; here n is the number of sites after deleting individually infeasible ones) and have redesigned the study (every table was regenerated):

- **Verification** (Section 5.1): every algorithm is checked against exhaustive search, including all $n!$ dispatch orders for $n \leq 7$ (400 instances), best-feasible-subset enumeration for $n \leq 14$ (1,500 instances, one third with weights up to 10^6 , FPTAS guarantee checked in 4,500 runs of which 2,923 had active scaling), and exhaustive search over assignments for the equal-cost algorithm with $m = 2, 3$ vehicles; the added results are checked by a further script (1,600 instances for the different-speed theorem, 600 for Proposition 4, 4,400 random instances plus an adversary for the refined FPTAS bound, and recursions cell by cell). No violations.
- **Accuracy across sizes** (Section 5.2): n up to 400, 50 instances per size, worst-case ratios as well as means, and a stronger baseline, EDD with skipping, in addition to naive EDD.
- **FPTAS with active scaling** (Section 5.3): weights up to 10^6 (K in the hundreds to thousands), a strongly correlated family, and an adversarial family. On random and strongly correlated instances the FPTAS is nearly exact (worst ratio 0.9993 at $\epsilon = 0.5$) because the bound is pessimistic when the optimum contains many sites; on the adversarial family the ratio falls to 0.669 at $\epsilon = 0.5$ and follows the guarantee. We now state that the adversarial column is a single deterministic instance equal to a closed form, not an empirical worst case. A simple completion step repairs the family completely.
- **Runtime** (Section 5.4) re-run with active scaling, and a **comparison with a general-purpose solver** (Section 5.5): HiGHS on the subset formulation returns the same optimum on all 40 instances and is 15 to 200 times slower than the dynamic program.

2 Reviewer 2

3. The part of MWHED's positioning in page 6 is comprehensible, nevertheless it could be rephrased better by shorter arguments.

Response. The “Positioning” paragraph (Section 2) was shortened and now lists the literatures it draws on and refers to Table 2; the two long paragraphs on real-time schedulability and matroid-partition algorithms were cut to two sentences and removed, respectively, since they carried no result used later.

1. Theorem 2 on NP-hardness of MWHED might require a bit efforts to get full clarity. In the 3rd line from the bottom of page 9, don't you think that the fact “completion times along a fixed set are additive” is too strong? What fixed sets should we use here? [...] The authors should explain about additivity of special sets [...] since it is not obvious by default, feasibility condition does not explicitly implies additivity. [...] the above condition ($\diamond$) $\sum_{i \in S} p_i \leq D$ is crucial to relate your formulation to a 0/1 knapsack instance.

Response. Thank you; the sentence was imprecise. We replaced it by an explicit statement, **Fact 1** (Section 4.1), with a proof: *if all deadlines equal D , a set S of sites can all be served on*

time iff $\sum_{i \in S} p_i \leq D$, which is exactly your condition ($\diamond$). The “fixed set” is an arbitrary subset S . The only additivity used is that the *last* site served from S completes at the sum of all of S ’s occupancies, because the vehicle is never idle and each p_i does not depend on the predecessor; this gives necessity. For sufficiency every completion time is a partial sum of positive numbers and so is at most the total. The proof of Theorem 3 cites Fact 1, and the knapsack correspondence for general weights is stated after the theorem. We also made the reduction handle odd total A and sites with $a_i > A/2$ explicitly.

2. [...] your claim on “NP-complete of the 0/1 knapsack problem” comes from polynomial reduction of PARTITION (one of the six popular graph-based NP-complete problems [...]). This is a solidly smart argument, if the above concern is clearly elucidated. Otherwise, a graph-based pure proof for Theorem 2 possibly might give better insights (ref. Delazeri and Ritt (2026)).

Response. We removed the dependence on the knapsack citation: Theorem 3 gives a *direct* polynomial reduction from PARTITION to MWHED ($p_i = w_i = a_i$, $d_i = A/2$), so that the proof is self-contained and shows hardness when $w_i = p_i$; membership in NP is added. A small remark on terminology: PARTITION is one of the basic NP-complete problems in Garey and Johnson, but it is a number problem, not a graph problem. We did not give a graph-based proof: MWHED has no graph structure once each visit is a round trip from the depot, so there is no natural graph on which to base one. Delazeri and Ritt (2026) prove NP-completeness for a different, graph-based model of wildfire suppression with simultaneous resources; Section 2 says so.

3. [...] the authors would elucidate more explicitly Theorem 5’s proof. In particular, would you remind readers the concepts (a) transversal matroid, union-find structure, and (b) the related notation $\alpha(n)$?

Response. Section 4.4 opens with a paragraph recalling the notions used: matroid and its three axioms, the greedy algorithm, the transversal matroid (sites against (vehicle, position) slots), and the union–find structure, with α defined as the inverse Ackermann function. The proof of Theorem 8 (5) is in five steps: (1) slots; (2) a counting criterion for feasibility; (3) a direct proof of the matroid augmentation property; (4) a short proof that greedy is optimal on any matroid; (5) the implementation, with a proof that it accepts a site exactly when the kept set plus that site is feasible. The theorem now allows vehicles with different dispatch times and is stated as classical. In answering this comment we also corrected our union–find remark: the $O(n\alpha(n))$ bound needs union by size or rank in addition to path compression (path compression alone gives $O(\log n)$ amortized), and we describe the label array that gives it for our linking direction. Algorithm 3 and Example 5 are rewritten accordingly.

4. [...] do we agree that the problem of MWHED is obviously data-driven, the proofs and optimal solutions obtained depend on both accurate formulation and precise calibration of model parameters, upon uncertainty? To some extent, would formulation of uncertainty [...] (at not just variable level but also at parameter one) be considered in next studies, where stochastic programming might be a potential approach? Kindly, e.g. refer to problems of evacuation planning under unsafe contexts and facility location under disruptive events (FLDE), ref. [1, 2].

Response. We agree. Section 6.3 (“Uncertain parameters”) covers uncertainty in p_i , d_i and w_i , not only in the deadlines. It discusses (i) scenario-based and two-stage stochastic formulations

and which of our results survive (the earliest-deadline lemma holds scenario by scenario but not simultaneously; a worked example shows that sorting by expected deadline can be nearly twice as bad as optimal), citing your two references (Wang, 2020; Koca, 2023); (ii) a robust-feasibility observation; and (iii) calibration and sensitivity analysis. The case study (Section 5.7) now includes a Monte-Carlo sensitivity analysis over response speed, documented event times and departure delay, in an independent and in a correlated variant, with standard errors: the plan that is optimal for the nominal instance is fully on time in only 16.2% of 20,000 draws and retains 31.9% of the drawn optimum on average, whereas a plan chosen against a sample of perturbed scenarios is fully on time in 79.3% and retains 99.7%. Stochastic programming for MWHED itself is listed as the main direction for future work; we have not attempted a solution method for it here.

3 Errors of our own corrected in this revision

In the course of revising we re-checked the manuscript, the code, the cited sources and the proofs, and found the following, which we correct and disclose.

1. **Case study: hazard times.** Two of the four hazard-arrival times had been *extrapolated* by an isotropic spread-rate rule (Magalia at 58 and Yankee Hill at 64 minutes), although the cited NIST report documents events for them: spot fires in Old Magalia at 08:40 and the fire running through Yankee Hill on the following day. The extrapolation was far off. We also had attributed a “time zero” of the timeline to NIST, but the report uses clock times only; the earlier anchors of 52 and 71 minutes were reproducible only with a start time taken from another source. The case study now uses the documented clock times of the first events in the four communities, with time zero equal to the initial dispatch time in NIST’s dispatch log, states the event definitions as NIST words them, and no longer attributes our ignition coordinate to NIST (the figure uses NIST’s origin point). All case-study results were recomputed; the earlier numbers (26,218, 26,928) are superseded.
2. **Case study: modelling premise.** Concow, Paradise and Yankee Hill are within about 10 km of each other but 22–37 km from the depot, so the round-trip model of the paper was far from the geometry, and the earlier conclusion that only Paradise could be protected was an artefact of that premise. The paper now proves that the round-trip model is conservative (Proposition 1), compares it with a chained route on the same data (Table 12), and says that the case study illustrates the model and does not support operational recommendations.
3. **Case study: protocol.** The “retained share” of the nominal plan was not reproducible from the text (an independent replication under another reading of the plan gave about 22% instead of 26.6%). The protocol (fixed visiting sequence, skipping sites deleted by Assumption 1) is now stated, the alternative reading is reported, standard errors are given, a correlated-error variant is added, and the weight at which Paradise would leave the optimum is reported. We also added a paragraph on the ethics of conceding low-weight sites and on what “protected” means.
4. **Case study: deadline semantics** (from the first revision). The model’s on-time test counts the completion of the round trip (return reading), but the narrative speaks of reaching the site (arrival reading). Remark 1 makes the two readings precise (the arrival reading is an instance of MWHED with deadlines $d_i^{\text{haz}} + p_i/2$) and the case study uses it; the return reading is reported for comparison. Individually infeasible sites are deleted before every algorithm, including the

naive baseline; statements that the optimum “saves” a number of people were replaced by the on-time weight.

5. **Novelty of Theorem 8 (5).** We had presented the extension of the equal-cost greedy to several identical vehicles as new. It is classical (Lawler 1976 for identical machines; Dessouky, Lageweg, Lenstra and van de Velde 1990 for uniform machines); we now cite both, state the theorem for vehicles with different dispatch times, and keep our proof for self-containedness. The classification (Theorem 12) is labelled an elementary corollary.
6. **Release dates.** The statements that equal-cost tractability “is not robust to release dates” and that hardness follows “once release dates are added” were not supported for one vehicle: the cited result of Heeger and Molter (2025) concerns the unweighted problem on several identical machines, and for one machine equal processing times with release dates and weights are polynomial (Baptiste, 1999). The text now says exactly this. The earlier inference that the equal-cost case becomes hard with several vehicles was also wrong, since that result requires release dates, which MWHED does not have.
7. **Tightness of the FPTAS.** The submitted text claimed that the guarantee $1 - \epsilon$ is “tight up to a factor $1 - \epsilon^2$ ”. The scheme in fact guarantees a larger constant $\rho_{n,\epsilon}$ (Theorem 7), which is attained by two families (Proposition 11); the classical $1 - \epsilon$ is not attained. The statements in the abstract, introduction, Table 2 and conclusion were rewritten.
8. **Proofs.** The main-text proof of the FPTAS guarantee used an inequality in the wrong direction (the appendix repaired it); the displayed recursions for the dynamic programs, written as an indicator times a value, were wrong as literally stated (they return 0 when the test fails), and are now written with cases; the proof of Lemma 2 did not cover ties between deadlines; the union-find cost claim needed union by size (see Reviewer 2, point 3); the reduction of Theorem 3 did not handle odd A explicitly. Each fix was checked against brute force (`verify_extensions.py`).
9. **Heeger–Hermelin.** The submitted text said that a small number of distinct dispatch costs “does not guarantee tractability”. The result is $W[1]$ -hardness in that parameter (no $f(k)\text{poly}(n)$ algorithm); the text now says this and makes no claim about two or more distinct values.
10. **Multiple vehicles.** The claim that the $m = 1$ instance “embeds” by adding idle vehicles was wrong (extra vehicles can serve sites); we now use blocker sites.
11. **Implementation.** The FPTAS guarantee relies on Assumption 1 (so that $w_{\max} \leq W^*$), which the code did not enforce; both algorithms now delete infeasible sites first, and the code was rewritten and checked against exhaustive search.
12. **Bibliography.** We verified every entry against a publisher, Crossref or arXiv record and corrected: the author lists and volume of Alvelos et al. (2025) (three authors, volume 86), the surname *von der Brüggen* (Günzel et al.), the author order of the NIST report, the article type of the AAMAS 2026 entry (extended abstract), and missing volume, article and page data for several proceedings entries.
13. **Verification claim.** The Lean 4 statement in our first draft of the Data Availability section was inconsistent about what is formalised (complexity-class and running-time statements are not); it now says precisely what is and is not formalised, and the library, the code and the experiments are public.
14. **Minor:** the dash convention in the dynamic-program example table was described incorrectly;

the explanation of why the FPTAS runtime was flat in ϵ cited saturating weights, whereas the cause was $K = 1$. Both are fixed.

We believe the revised manuscript is considerably more honest about what is and is not new, and we thank the reviewers for the comments that led to these changes.

Sincerely,
The authors

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